# Use Case Specification: UC-21

- **ID / Code:** UC-21
- **Use Case Name:** Fetch Bank Transactions
- **Description:** Subfunction included by UC-20 that connects to the external Banking API (or simulated Stripe payment environment) to retrieve new incoming transactions since the last successful synchronization (FR-3.1, FR-3.4).
- **Primary Actor:** None (included use case). Supporting actor: Banking API
- **Scope:** Subscription Management & Invoicing System
- **Level:** Subfunction
- **Stakeholders & Interests:**
  - *Company (Product Owner):* Wants reliable, complete transaction data from banking APIs or test payment environments (FR-3.1, FR-3.4, NFR-2.2).
  - *Bank / Payment Provider:* Requires secure, authenticated API usage.
- **Preconditions:** Bank integration or simulated Stripe credentials and settings are configured (FR-3.1, FR-3.4, UC-28).
- **Postconditions:** New transactions are stored for reconciliation and the synchronization checkpoint is updated.
- **Success Guarantee:** All transactions since the last checkpoint are retrieved and stored without loss or duplication (NFR-2.2).
- **Minimal Guarantee:** If retrieval fails, the checkpoint is not advanced, so no transaction is lost.
- **Trigger:** Invocation from UC-20.
- **Main Success Scenario:**

1. **System:** Requests from the **Banking API** (or simulated Stripe service, FR-3.4) the transactions since the last synchronization checkpoint.
2. **Banking API:** Returns the list of transactions.
3. **System:** Validates the response format and discards transactions already stored.
4. **System:** Stores the new transactions in "pending reconciliation" state and updates the checkpoint.
5. **System:** Returns control to UC-20.

- **Extensions:**
  - 2a. Banking API or payment service unavailable, timeout or authentication error:
    - 2a1. System retries according to the configured policy.
    - 2a2. If it keeps failing, System logs the error, keeps the checkpoint unchanged and reports the failure to UC-20.
  - 3a. Malformed or incomplete response:
    - 3a1. System rejects the batch, logs it and reports an integration error.